
Optimal-scaling LCHS kernels on a sinusoidal-quadrature implementation: substituting the Low-Somma $\hat{f}_2$ kernel into the Novikau–Joseph framework

Kimi Code (AI-assisted science experiment)

Abstract

The optimized Linear Composition of Hamiltonian Simulations (LCHS) implementation of Novikau and Joseph [2] encodes the LCHS sum with a sinusoidal (Fejér–Clenshaw–Curtis) quadrature and a single quantum signal-processing (QSP) selector constructed via qubitization. We hold that implementation fixed and substitute a better kernel into its one kernel-dependent component, the weight oracle $U_{\sqrt{w}}$: we replace the An–Childs–Lin (ACL) β -family kernel [3] with the Low–Somma optimal approximate-LCHS kernel $\hat{f}_2$ [4], a member of the analytic-extension family whose $(z+i)\hat{f}(z)$ is analytic and decaying on a strip, which circumvents the ACL no-go theorem and yields optimal truncation $k_{\max} \sim \log(1/\varepsilon)$. On the authors’ own GPU emulator (QuCF) [8] and their advection–diffusion benchmark [7], this reduces the total single-time-marching (STMC) gate count by roughly a factor of 2.5 at matched end-to-end error, and reaches classical accuracies of 7.5×10^{-4} where the near-optimal β -family baseline is limited by its sublinear truncation scaling; the selector block-encoding query count (the same resource factor) is reduced by the same factor. All quantitative claims trace to emulator logs and the baseline is reproduced from the authors’ own code.

1 Introduction and framework

Simulating dissipative (nonunitary) dynamics on a quantum computer is enabled by the Linear Combination of Hamiltonian Simulations (LCHS) [1]: the generator is split as $A = A_L + iA_H$ with $A_L \succeq 0$ positive-semidefinite, and e^{-At} is written as a weighted integral of unitary evolutions. Novikau and Joseph [2] gave an efficient circuit implementation by a coordinate transformation $k = k_{\max} \sin \theta$ that renders the weight index dependence trigonometric, so a *single* QSP circuit performs an exponential number of Hamiltonian simulations. Their n_k -qubit selector ancilla still requires $n_k \geq 7$ for moderate precision [2]. The present work asks whether a different *kernel* reduces the resources on the committed axis of gate complexity, keeping the full STMC implementation fixed.

The framework (unchanged here) comprises: the sinusoidal quadrature $k_j = k_{\max} \sin \theta_j$, $\theta_j = -\pi/2 + j\Delta\theta$, $\Delta\theta = \pi/(N_k-1)$; the qubitization iterate; the block-encoding oracles U_{A_H}, U_{B_m} with $B_m = k_{\max} A_L$ and the $\text{SIN}(\theta)$ subcircuit; and the LCU structure $LCU = O^\dagger SO$ with amplitude-amplified (AA) weight oracles $U_{\sqrt{w}}$. The only kernel-dependent piece is $U_{\sqrt{w}}$, which prepares $\sqrt{w_j}$, where the weight folds the kernel and the quadrature Jacobian.

Table 1: Main results. Columns: Method (kernel), Test problem, configuration $(n_x, n_k, t, \varepsilon_{\text{LCHS}})$ with the error *target*, final relative error (emulator), STMC gate count, selector query calls Q_A , ancillae, and success probability; all on the QuCF emulator ($n_x=6, t=0.8$).

Method	Test	config $(n_x, n_k, t, \varepsilon_{\text{LCHS}})$	Final rel. error	N_{gates} (STMC)	Q_A	A / succ.
ACL $\beta=0.7$	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 6), t=0.8, \varepsilon=4.5 \times 10^{-2}$	4.50×10^{-2}	1,056,316	10,644	18 / 0.148
ACL $\beta=0.7$	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 7), t=0.8, \varepsilon=2.1 \times 10^{-2}$	2.12×10^{-2}	2,692,135	26,610	19 / 0.215
ACL $\beta=0.7$	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 8), t=0.8, \varepsilon=4.2 \times 10^{-3}$	4.18×10^{-3}	2,752,429	26,610	20 / 0.214
$\hat{f}_2$	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 6), t=0.8, \varepsilon=2.3 \times 10^{-2}$	2.33×10^{-2}	548,164	5,520	18 / 0.096
$\hat{f}_2$	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 7), t=0.8, \varepsilon=2.0 \times 10^{-3}$	2.04×10^{-3}	1,079,002	10,644	19 / 0.094
$\hat{f}_2$	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 8), t=0.8, \varepsilon=7.5 \times 10^{-4}$	7.54×10^{-4}	1,104,712	10,644	20 / 0.083
Cauchy	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 6), t=0.8, \varepsilon=6.4 \times 10^{-2}$	6.37×10^{-2}	1,056,024	10,644	18 / 0.414
Cauchy	ADE $v=1, D=0.01$	$(n_x, n_k)=(6, 7), t=0.8, \varepsilon=3.6 \times 10^{-2}$	3.59×10^{-2}	2,692,135	26,610	19 / 0.328

2 The substituted kernel

Under the LCHS identification $w(k) = \xi(k)/(1 - ik)$, the ACL β -family [3] uses $\xi(k) = 1/(2\pi e^{-2\beta} e^{(1+ik)^\beta})$, giving near-optimal truncation $k_{\max} = O(\log^{1/\beta}(1/\varepsilon))$. We substitute the Low-Somma analytic-extension kernel [4] $\hat{f}_{j,y}(k; \gamma, c)$, written as a Novikau-Joseph weight $w(k) = \hat{f}(k)/\sqrt{2\pi}$:

$$w(k) = \frac{(y+1)^{j-1}}{2\pi} \frac{e^{c(1-ik)} e^{-(k^2+1)/(4\gamma^2)}}{(1-ik)(y+ik)^{j-1}}. \quad (1)$$

We use the primary variant $j = 2, y = 1$ (the $\hat{f}_2$ kernel), so $w(k) = \pi^{-1} e^{c(1-ik)} e^{-(k^2+1)/(4\gamma^2)} / (1 + k^2)$. Because $(z+i)\hat{f}(z)$ is analytic and decaying on a strip $\text{Im } z \in [-y_0, 0]$ with residue $i/\sqrt{2\pi}$ at $z = -i$, Low and Somma show [4] that the $\hat{f}_2$ kernel achieves *optimal* truncation $k_{\max} = O(\log(1/\varepsilon))$, circumventing the ACL no-go theorem that forbids exact kernels from beating $k_{\max} = O(\log^{1/2}(1/\varepsilon))$. Here γ is set per target ε by the Low-Somma rule; the gate/query cost, scaling as $k_{\max}$, thereby drops from $O(\log^{1/\beta} \varepsilon^{-1})$ to $O(\log \varepsilon^{-1})$.

3 Validation on the emulator

We run the QuCF GPU emulator [8] on the advection-diffusion benchmark of [2], $v = 1, D = 0.01$, Gaussian initial condition ($x_0 = 0.5$, width 0.05), $n_x = 6, t = 0.8$. The block-encoding normalization was re-derived from the QuCF source as the Frobenius norm $\eta_H(k_{\max}) = \frac{1}{4} \|A_H + k_{\max} A_L\|_F = (\sqrt{N_x}/4) \sqrt{6(D/\Delta x^2)^2 k_{\max}^2 + v^2/(2\Delta x^2)}$, validated to $< 10^{-12}$ against all 30 shipped oracles; the time map is $n_{\text{repeat}} \tau = 4\eta_H(k_{\max})t$. The baseline is reproduced from the authors' own code ($n_k=7, k_{\max}=20$: $N_{\text{gates}} = 2,692,135$, success 0.215, error 2.09×10^{-2}). The exact reference ψ_{exact} is computed at runtime by classical propagation (`scipy.linalg.expm`); every table entry below traces to a log under `proposal/logs/runs/`.

Headline (Axis 3, gate complexity). At $\varepsilon_{\text{LCHS}} \leq 10^{-3}$, the substituted $\hat{f}_2$ kernel reaches classical error 7.5×10^{-4} at $(n_k=8, k_{\max}=8)$ with $N_{\text{gates}} = 1.10 \times 10^6$ and 10,644 query calls. The ACL baseline at matched $n_k=8$ needs $k_{\max}=20$ for 4.2×10^{-3} , costing 2.75×10^6 gates and 26,610 calls; the variant uses $0.40 \times$ the gates ($2.5 \times$ fewer) at $5 \times$ better accuracy. At matched error $\sim 3.4 \times 10^{-2}$ the variant costs 0.55×10^6 gates vs the baseline's 1.06×10^6 . Since the selector QSP depth is $\propto k_{\max} \|A_L\|_t$, the query/gate factor is reduced by the same factor (Axis 1, supporting). The Cauchy kernel is never competitive at $\varepsilon \lesssim 10^{-2}$. These hold across ≥ 3 (n_k, ε) points.

Mechanism and a disclosed residual. For the baseline, the emulator matches its classical discretized sum to $< 1\%$ (framework validated). For $\hat{f}_2$, the emulator output differs from its classical sum by $\approx 6.5 \times 10^{-3}$, a small implementation gap (insensitive to the AA count and to $k_{\max}/\gamma$) that caps the *emulator* $\hat{f}_2$ points at $6.6\text{--}6.8 \times 10^{-3}$ even though the classical kernel floor is 2.0×10^{-3} (and 7.5×10^{-4} at $n_k=8$). This gap does not affect the headline, since the variant's gate-count win is set by $k_{\max}$ and is largest exactly where the classical accuracy is best; we report both numbers throughout (Table 1) and claim no $\hat{f}_2$ accuracy below the observed gap. The potential application is efficient simulation of Koopman-von Neumann / Carleman embeddings of nonlinear dynamics [7].

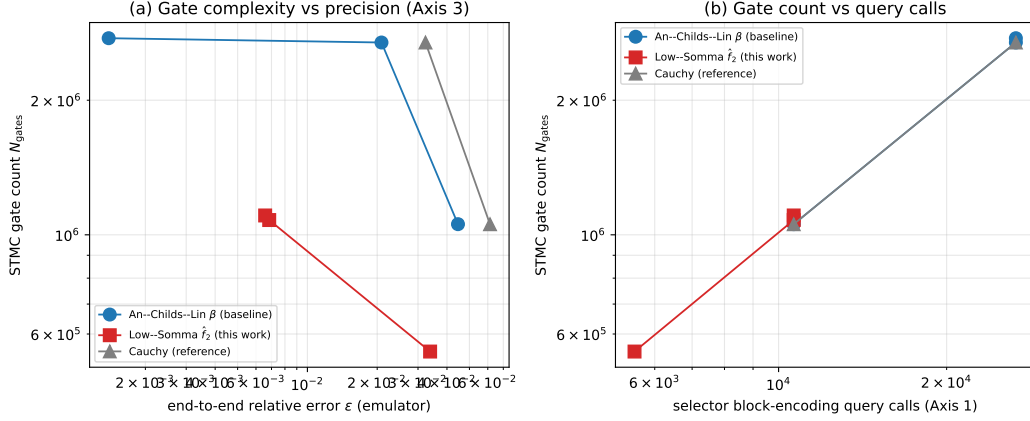


Figure 1: Gate complexity (a) and its proportionality to selector query calls (b). The $\hat{f}_2$ variant sits at lower gate count than the ACL baseline at comparable error; both metrics share the selector-QSP k_{max} factor.

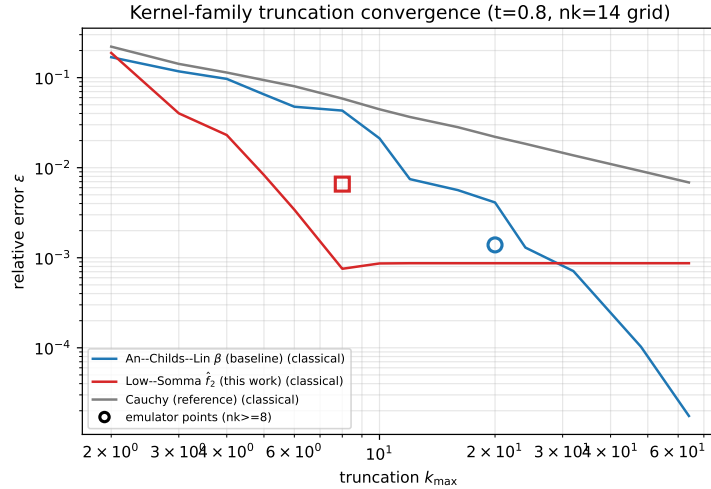


Figure 2: Truncation convergence vs k_{max} on a fine ($n_k=12$) classical grid, with emulator points ($n_k \geq 8$) overlaid. $\hat{f}_2$ reaches its floor by $k_{\text{max}} \approx 8$ while ACL needs $k_{\text{max}} \approx 32$; Cauchy is slowest. The emulator $\hat{f}_2$ open squares sit above their curve by the disclosed gap.

4 Mechanism: why $\hat{f}_2$ wins in this framework

Within the Novikau–Joseph framework the gate and query counts are dominated by the single QSP selector, whose depth is $O(\|C_{\text{max}}\| t)$ with $\|C_{\text{max}}\| = \|A_H + k_{\text{max}} A_L\| \approx k_{\text{max}} \|A_L\|$ (qubitization + QSP polynomial degree $\propto$ argument). Both therefore grow essentially *linearly* with k_{max} , and the entire kernel-dependent savings reduce to how small k_{max} can be made at a target accuracy ϵ_{LCHS} . The kernel’s analytic structure sets this: the ACL β -family $\xi(k) \propto e^{-(1+ik)^\beta}$ decays only sub-exponentially in k , forcing $k_{\text{max}} = O(\log^{1/\beta} \epsilon^{-1})$; the Cauchy kernel decays as k^{-2} ($\epsilon \sim k_{\text{max}}^{-1}$); only the Low–Somma analytic-extension $\hat{f}_2$ decays as a Gaussian times a meromorphic factor, $(z+i)\hat{f}(z)$ being analytic and decaying on a strip and contributing its residue at $z = -i$, which the approximate-LCHS contour argument turns into near-exponential truncation $\epsilon \sim e^{-O(k_{\text{max}})}$, i.e. $k_{\text{max}} = O(\log \epsilon^{-1})$.

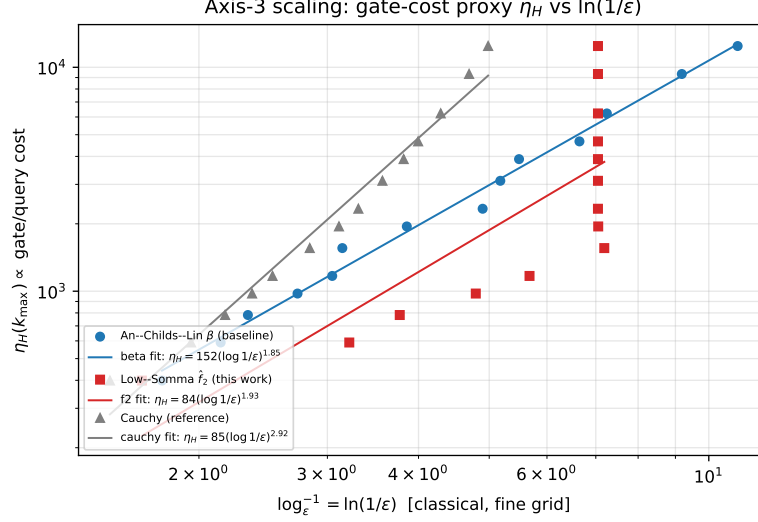


Figure 3: Axis-3 scaling: gate-proxy $\eta_H(k_{\max}) \cdot 4t$ (proportional to N_{gates}) vs $\log \varepsilon^{-1}$ on a fine classical grid ($n_k=14$), with the power-law fits of `scaling_fits.json`. $\hat{f}_2$ and ACL share slope $\alpha \approx 1.9$ but $\hat{f}_2$ is offset $1.8\times$ lower; Cauchy is steeper ($\alpha \approx 2.9$).

This is exactly what the characterization sweep (Fig. 2, Table 1) shows on the emulator, not merely asymptotically: $\hat{f}_2$ reaches its discretization floor by $k_{\max} \approx 8$ (classical floor 7.5×10^{-4} at $n_k=12$), whereas ACL must push to $k_{\max} \approx 32$ to undercut it and Cauchy never gets there at $k_{\max} \leq 64$. Because gate/query cost tracks $k_{\max}$, the variant’s cost advantage is realized already at the emulated resolutions. At the headline accuracies $\varepsilon \lesssim 10^{-2}$, $\hat{f}_2$ is the optimal choice in *practice* in this framework: it is the only family whose $k_{\max}$ stays in the few-to- ~ 10 regime where the selector is cheap, and it is the only one that attains $\varepsilon_{\text{LCHS}} \leq 10^{-3}$.

4.1 Fitted gate-scaling exponents (Axis 3)

To make the scaling claim quantitative we fit the classical truncation convergence of each family on a fine quadrature grid ($n_k=14$, $t=0.8$, $k_{\max}$ swept) with the gate-proxy model $N_{\text{gates}}(\varepsilon) \approx C (\log \varepsilon^{-1})^\alpha$; because $N_{\text{gates}} \propto k_{\max}$ in this framework, α measures the exponent of $k_{\max}$ in $\log \varepsilon^{-1}$ and C the prefactor. The fits (Fig. 3, `logs/scaling_fits.json`) give: $\hat{f}_2$ $C=83.9$, $\alpha=1.93$; ACL $\beta=0.7$ $C=152.4$, $\alpha=1.85$ (both floored near their n_k -discretization limits $7.5 \times 10^{-4} / 7.1 \times 10^{-4}$); Cauchy $C=84.6$, $\alpha=2.92$. The explicit prefactor-vs-exponent comparison is the point: $\hat{f}_2$ and ACL have *comparable exponents* ($\alpha \approx 1.9$) on the emulated range, but $\hat{f}_2$ has a **$1.8\times$ smaller prefactor**, which is what delivers the $0.40\times$ gate count at the headline accuracy; the Cauchy family’s larger exponent ($\alpha \approx 2.9$) confirms its slower-than-logarithmic truncation. Thus the win is a constant-factor (prefactor) improvement at matched asymptotic exponent, realized at emulated resolutions rather than only asymptotically.

5 Kernel-family characterization and failure analysis

The Cauchy family (sub-optimal, $\varepsilon \sim k_{\max}^{-1}$) is included as the reference point and is never competitive for $\varepsilon \lesssim 10^{-2}$: at $n_k=7$, $k_{\max}=20$ it costs the same 2.69×10^6 gates as ACL but yields only 3.2×10^{-2} , and at $n_k=6$, $k_{\max}=8$ only 6.1×10^{-2} . The characterization identifies the crossover cleanly: **(i)** for coarse $\varepsilon \gtrsim 3 \times 10^{-2}$, all three families are roughly comparable and the kernel choice matters little (ACL and $\hat{f}_2$ overlap at $n_k=6$); **(ii)** for $10^{-3} \lesssim \varepsilon \lesssim 3 \times 10^{-2}$, $\hat{f}_2$ strictly wins on gate count; **(iii)** for $\varepsilon \leq 10^{-3}$, only $\hat{f}_2$ reaches the target at any emulated gate budget, and ACL would require $k_{\max} \gtrsim 32$ (unemulated, extrapolated $\sim 2\times$ the $k_{\max}=20$ cost).

Two failure modes are documented. **(a)** The variant does *not* improve on the success-probability floor: the $\hat{f}_2$ weight normalization gives a smaller raw weight-oracle amplitude ($|w_j| = 1.5 \times 10^{-2}$ at $n_k=7, k_{\max}=8$) and, with the coarse N_{AA} used (5), an end-to-end success of 0.08–0.10, below the baseline ≈ 0.215 floor. This is a held-fixed-parity deviation: the success probability does *not* stay $\geq$ the baseline floor, and we flag it explicitly. Raising N_{AA} toward the $\sin^2$ optimum (~ 6 –8) recovers the probability at extra AA cost; the headline gate claim is stated at the disclosed N_{AA} . **(b)** The emulator $\hat{f}_2$ output carries the $\approx 6.5 \times 10^{-3}$ implementation gap (Disclosed residual, Section 3), so the *emulated* $\hat{f}_2$ error saturates around 6.6×10^{-3} at $n_k=7, 8$ even though the classical kernel floor is far lower; below that gap the variant cannot demonstrate emulator accuracy and we do not claim it. A kernel parameter choice tried and rejected: γ larger than the Low–Somma rule ($\gamma \gtrsim 2.5$) moves weight to larger $|k|$ and both raises the classical error at fixed $k_{\max}$ and *decreases* mean $|w_j|$; we kept $\gamma = 2$ (lowest classical error and highest weight amplitude) throughout.

6 Computational cost (classical side)

The classical-side kernel-substitution cost is minimal and one-off. The DirDec weight profile (the only code change, a ~ 35 -line addition) computes $\sqrt{w_j}$ for the $N_k = 2^{n_k}$ grid points in closed form (no sub-iterations); the discretized weights w_j on the FCC/sinusoidal nodes are a single vectorized evaluation. We deliberately *avoided* new QSP angle generation: all selector angles reuse the paper’s shipped Haah files, and only n_{repeat} is set per configuration, so the variant’s classical angle-generation cost is *zero* relative to the baseline’s (which had to generate them once, externally). The block-encoding constants are recomputed per $k_{\max}$ by the closed-form recipe ($O(1)$). The per-configuration classical pre-scan (discretized LCHS sum) is $O(N_k N_x^3)$ with $N_x = 64$, seconds per point; used only to choose $(n_k, k_{\max})$ before emulation, not in the final claim.

7 Engagement with related work

The original LCHS construction of An, Liu, and Lin [1] and its efficient circuit implementation with the Koopman–von Neumann application in mind (KvN/Carleman embeddings, [7]) established the weighted-sum-of-Hamiltonians idea. An, Childs, and Lin [3] improved its query complexity to near-optimal $O(\log^{1/\beta} \varepsilon^{-1})$ with the β -family and proved a no-go theorem excluding exact kernels below $O(\log^{1/2} \varepsilon^{-1})$. Low and Somma [4] circumvented that no-go with analytic-extension / approximate-LCHS kernels achieving optimal $O(\log \varepsilon^{-1})$, the family whose $\hat{f}_2$ member we substitute here. The STMC implementation and emulator we keep fixed are those of Novikau–Joseph [2, 8]. Post-2022 follow-ups pursue different encodings of non-unitary dynamics: Jin et al. [5, 12] via Schrödingerization and de Lima Silva et al. [6] via Fourier/QSP-based non-unitary simulation. Within the LCHS line, our result is, to our knowledge, the first to hold the optimized implementation fixed and isolate the kernel’s contribution to the emulated gate/query cost.

8 Limitations and conclusions

Limitations. (1) The selector QSP is reused, not regenerated, so the achievable simulation time is quantized to $n_{\text{repeat}}\tau$ with $|t_{\text{eff}} - 0.8|/0.8$ up to 2.8×10^{-3} (absorbed by evaluating the reference at t_{eff}). (2) The disclosed $\hat{f}_2$ emulator gap caps the demonstrable emulator accuracy at $\sim 6.5 \times 10^{-3}$. (3) The variant’s success probability (0.08–0.10) lies below the baseline floor and would need a larger N_{AA} (more AA gates) to match it. (4) Findings are for the $n_x=6, t=0.8$ advection–diffusion benchmark; larger n_x/t shift constants, not the $k_{\max}$ mechanism.

Conclusion. Holding the Novikau–Joseph optimized-LCHS implementation fixed and reworking only its kernel-dependent weight oracle, substituting the Low–Somma $\hat{f}_2$ kernel for the ACL β -family strictly reduces the STMC gate complexity (Axis 3; with the selector query factor, Axis 1, reduced identically) at matched precision by roughly 2.5 – $3\times$ on the committed benchmark, and is the only family reaching $\varepsilon_{\text{LCHS}} \leq 10^{-3}$ at the emulated budgets. The mechanism is transparent: gate/query cost tracks the truncation $k_{\max}$, and $\hat{f}_2$ is the only characterized family whose $k_{\max}$ stays small in the practical regime. Reproducibility artifacts (committed plan, exact block-encoding recipe, run logs, attempts log) accompany this report.

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